

Effective versions of equilogical spaces

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Abstract

The full category of equilogical spaces can be thought of either as $ER(TOP)$, the category of equivalence relations on T_0 -topological spaces and equivariant continuous maps (i.e., equivalence preserving), or $PER(DOM/T)$, the category of partial equivalence relations on Scott domains with a top (i.e., algebraic lattices) and equivariant continuous maps, or $PER(POWER)$, the category of partial equivalence relations on power-set spaces and equivariant continuous maps. (See also the the abstract of Andrej Bauer at this meeting for a discussion of another equivalent category.) If we replace general topological spaces with countably based spaces - each given with a particular choice of a countable basis - we can then ask how the resulting categories express effective (computable) notions. The most explicit equivalent version of the category obtained is $PER(P(N))$, the category of partial equivalence relations over the power-set-of- N domain and equivariant continuous maps.

The talk will discuss how the notion of computability appears in this category and how the category can be used as a model for a type theory enriched with a 'computability' modal operator. (See also the abstract of Lars Birkedal at this meeting.)
